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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBTG300, Discipline Specific Elective (DSE)- MG5DSEBBT304

RECOMBINANT DNA TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Questions-Answer any 10 questions

Each question carries 2 marks

1. Name any two antibiotics that belongs to the class of aminoglycosides. [U] / [CO2]
   Any two- 2 marks
2. State the purpose of using adapters in molecular cloning. [U] / [CO1]
   Provide compatible ends- 2 marks
3. Mention two applications of RAPD. [U] / [CO3]
   Any two-Genetic diversity analysis/identification of varieties or strains etc- 2 marks
4. What is the tet regulatory system? [U] / [CO2]
   A gene regulatory system based on tetracycline- responsive regulatory elements that allows controlled transgene expression- 2 marks.
5. What is meant by primer extension? [U] / [CO3]
   DNA strand synthesis- 2 marks
6. Name the vector used for gene transfer in both bacteria and plants. [K] / [CO1]
   pCAMBIA
7. Define In-Vivo gene therapy. [K] / [CO3]
   Direct delivery of genes into the cells- 2 marks.
8. Which gene is used in blue-white screening? [U] / [CO2]
   lacZ gene – 2 marks
9. Name any two qualitative agronomic traits improved through rDNA technology. [K] / [CO3]
   Any two- Herbicide resistance/insect resitance etc- 2 marks
10. What information can be obtained from a footprinting experiment? [An] / [CO3]
    The protein binds to a specific DNA region and protects it from enzymatic cleavage, producing a detectable footprint-2 marks.

11. Name the process of assembling bacteriophage particles outside a living cell. [U] / [CO1]
    In vitro packaging- 2 marks
12. What is the major difference between electroporation and microinjection?. [K] / [CO2]
    Electroporation uses electrical pulses to create temporary pores whereas microinjection directly injects DNA into a cell using micropipette.- 2 marks [2x10 = 20]

Part B

Short Essay Questions-Answer any 4 questions

Each question carries 5 marks

13. Explain the significance of the intergenic region and cloning site in M 13 vector construction. [U] / [CO1]
    M 13 vector construction- introduction- 1 mark, significance -intergenic region- contains the viral origin of replication and transcription terminators without disrupting essential genes, and cloning site- allows safe insertion of foreign DNA to yield single stranded DNA templates -4 marks
14. Discuss about the Hygromycin and Puromycin resistance in animals. [K] / [CO2]
    Genetic markers in animals- introduction- 1 mark, Hygromycin resistance- explain- 2 marks, Puromycin resistance- explain- 2 marks.
15. Write a short note on recombinant insulin. [U] / [CO3]
    Recombinant insulin- introduction- 1 mark, production- steps- 3 marks, significance- 1 mark.
16. Describe the characteristic features of pSC vectors. [U] / [CO1]
    Features of pSC vectors- 5 points-5 marks
17. Discuss the importance of affinity tags in downstream processing of recombinant proteins. [U] / [CO2]
    Recombinant proteins -downstream processing-introduction- recovery and purification-2 marks , importance of affinity tags-3 marks
18. How do restriction endonucleases facilitate DNA manipulation in genetic engineering? [U] / [CO1]
    Restriction endonucleases- introduction- 1 mark, applications in genetic engineering- role in genetic manipulation-cutting DNA at specific sites, creating sticky or blunt ends, constructing rDNA, etc- 4 marks
    [5x4 = 20]

Part C

Essay Questions-Answer any 1 questions

Each question carries 10 marks

19. Describe the organization of a typical expression cassette and explain the functions of its major components from promoter to transcription termination. [U] / [CO2]
    An expression cassette-definition- **2 marks**, organization of a typical expression cassette-5'→ [promoter] →[5'UTR/Kozak]→[gene of interest(ORF)→[3'UTR]→[polyadenylation signal]→3'-**3 marks**, functions of its major components- **5 marks**.
20. Discuss the major NGS platforms—Illumina, SOLiD, Ion Torrent and 454. [K] / [CO3]
    NGS- introduction- **1 mark**, advantages- **1 mark** explain major platforms- Illumina- **2 marks**, SOLiD- **2 marks,** Ion Torrent- **2 marks**, 454- **2 marks** [10x1 = 10]